

Special Techniques in Electrostatics

Problem of first kind:

Surface of the conductor defined the BC and the region of interest where field was examined contained a point charge

Problem of second kind

Potential/charge density is specified on the boundary of some region, and we are asked to find potential/field in the interior where there is no point charge

Almost in all such cases, boundary (conductor) surface have shapes that corresponds to constant coordinate surfaces in some CS

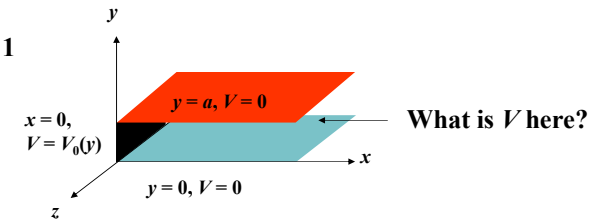
We can not use Method of Images, rather need to solve Laplace Equation!

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Special Techniques in Electrostatics

Method needed to solve Laplace Eq is known as **SEPARATION OF VARIABLES**

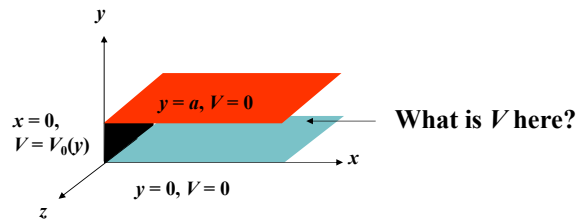
Example: 1



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Special Techniques in Electrostatics: Example 1

Method needed to solve Laplace Eq is known as **SEPARATION OF VARIABLES**



1. Look for symmetry

There is no charge in the region
Potential varies along x as well as along y
Potential does not vary along z at all

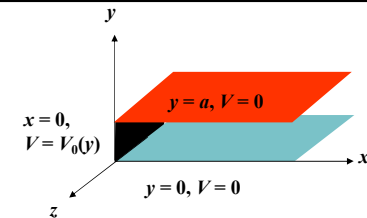
$$\longrightarrow V = V(x, y) \quad (1)$$

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2. Define BC

- $V = 0$, at $y = 0$
- $V = 0$, at $y = a$
- $V = V_0(y)$ at $x = 0$
- In addition: $V \rightarrow 0$ as $x \rightarrow \infty$ (not given but obvious)



3. Laplace Eq (Homogeneous DE)

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \quad (2)$$

4. Variable separable method

$$V(x, y) = X(x)Y(y) \quad (3)$$

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Laplace Eq becomes

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} = 0 \quad (4)$$

Dividing by (3) $V(x, y) = X(x)Y(y)$

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0 \quad (5)$$

$$f(x) + g(y) = 0$$

$$C_1 + C_2 = 0$$

$$\rightarrow \frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2 \quad \text{PDE} \rightarrow \text{ODE}$$

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Special Techniques in Electrostatics: Example 1

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2 \quad \text{PDE} \rightarrow \text{ODE}$$

Assume $C_2 = +ve$

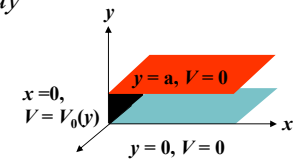
$$C_2 = k^2$$

$$\frac{1}{Y} \frac{d^2 Y}{dy^2} = k^2 \rightarrow \frac{d^2 Y}{dy^2} - k^2 Y = 0$$

Solution would be $e^{\pm ky}$

$$\rightarrow \text{Hence } C_1 = k^2 \quad C_2 = -k^2$$

$$\frac{d^2 X}{dx^2} - k^2 X = 0 \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0$$



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Special Techniques in Electrostatics: Example 1

$$\frac{d^2 X}{dx^2} - k^2 X = 0 \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0 \quad (6)$$

$$\rightarrow X(x) = Ae^{kx} + Be^{-kx}$$

$$Y(y) = C \sin ky + D \cos ky$$

$$\text{OR } V(x, y) = (Ae^{kx} + Be^{-kx}) (C \sin ky + D \cos ky) \quad (7)$$

Variable separable solution to the Laplace Equation

4. Application of Boundary Condition

iv. $V \rightarrow 0$ as $x \rightarrow \infty \rightarrow A \rightarrow 0$ Assuming k is +ve

$$V(x, y) = Be^{-kx} (C \sin ky + D \cos ky)$$

$$\text{OR } V(x, y) = e^{-kx} (C \sin ky + D \cos ky)$$

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4. Application of Boundary Condition

iv. $V \rightarrow 0$ as $x \rightarrow \infty \rightarrow A \rightarrow 0$

$$V(x, y) = e^{-kx} (C \sin ky + D \cos ky)$$

i. $V \rightarrow 0$ at $y \rightarrow 0 \rightarrow D \rightarrow 0$

$$V(x, y) = Ce^{-kx} \sin ky$$

ii. $V \rightarrow 0$ at $y \rightarrow a \rightarrow 0 = Ce^{-kx} \sin ka$

$$k = n\pi/a \quad n = 1, 2, 3, \dots$$

Note: k can not be -ve and can not be zero

Observe: BCs are applied in a way to first make all possible constants zero (in this case A and D).

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Solution to LE for this problem

$$V(x, y) = Ce^{-n\pi x/a} \sin(n\pi y/a) \quad n = 1, 2, 3 \dots$$

Infinite Set

General Solution to LE, which is Homogeneous DE, for this problem

$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin(n\pi y/a) \quad (8)$$

Application of Boundary Condition

iii. $V \rightarrow V_0(y)$ at $x \rightarrow 0$

$$V(0, y) = \sum_{n=1}^{\infty} C_n \sin(n\pi y/a) = V_0(y)$$

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Special Techniques in Electrostatics: Example 1

Consider an interval $[a, b]$ for variable x

Consider functions $U_n(x)$:

$U_1(x), U_2(x), \dots$ defined on $[a, b]$

$$\int_a^b U_n(x) U_m(x) dx = s_n \delta_{mn} \quad \text{Orthogonal functions}$$

if $s_n = 1$ **Orthonormal functions**

$$\int_a^b U_n(x) U_m(x) dx = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

$U_n(x): \{U_1(x), U_2(x), \dots\}$ form a complete set on $[a, b]$

$$\longrightarrow V(x) = \sum_{n=1}^{\infty} C_n U_n(x)$$

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Special Techniques in Electrostatics: Example 1

Example: $\sin\left(\frac{n\pi x}{a}\right)$ in an interval $[0, a]$, n a +ve integer

$$\int_0^a \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx = \frac{a}{2} \delta_{mn}$$

$\longrightarrow$ **Orthogonal functions**

$$\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \text{ in an interval } [0, a], n \text{ a +ve integer}$$

$\longrightarrow$ **Orthonormal functions**

We shall exploit this property of sin function

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Special Techniques in Electrostatics: Example 1

$$V(0, y) = \sum_{n=1}^{\infty} C_n \sin(n\pi y/a) = V_0(y)$$

$$\int_0^a \sum_{n=1}^{\infty} C_n \sin(n\pi y/a) \sin(n'\pi y/a) dy = \int_0^a V_0(y) \sin(n'\pi y/a) dy$$

$$\sum_{n=1}^{\infty} C_n \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy = \int_0^a V_0(y) \sin(n'\pi y/a) dy$$

$$\sum_{n=1}^{\infty} C_n \frac{a}{2} \delta_{nn'} = \int_0^a V_0(y) \sin(n'\pi y/a) dy$$

$$C_{n'} = \frac{2}{a} \int_0^a V_0(y) \sin(n'\pi y/a) dy$$

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Special Techniques in Electrostatics: Example 1

Unique solution to the given problem:

$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin(n\pi y/a)$$

$$C_n = \frac{2}{a} \int_0^a V_0(y) \sin(n\pi y/a) dy$$

OR $C_n = \frac{2}{a} \int_0^a V_0(y) \sin(n\pi y/a) dy$

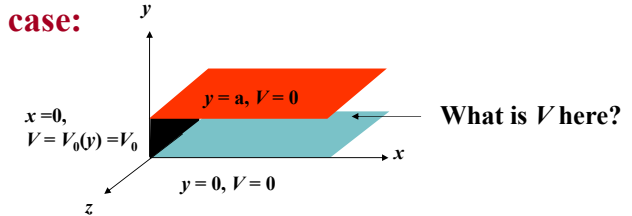
$$V(x, y) = \sum_{n=1}^{\infty} \left\{ \frac{2}{a} \int_0^a V_0(y) \sin\left(\frac{n\pi y}{a}\right) dy \right\} e^{-n\pi x/a} \sin(n\pi y/a)$$

This is the only solution to LE for this problem

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Special Techniques in Electrostatics: Example 1

Special case:



$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin(n\pi y/a)$$

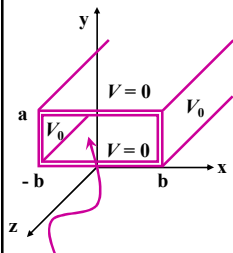
$$C_n = \frac{2}{a} \int_0^a V_0(y) \sin(n\pi y/a) dy = \frac{2V_0}{a} \int_0^a \sin(n\pi y/a) dy = \frac{2V_0}{a} \frac{a}{n\pi} \left[-\cos\left(\frac{n\pi y}{a}\right) \right]_0^a$$

$$= \frac{2V_0}{n\pi} (1 - \cos n\pi) = \begin{cases} 0 & n \text{ being even} \\ \frac{4V_0}{n\pi} & n \text{ being odd} \end{cases}$$

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n} e^{-n\pi x/a} \sin(n\pi y/a)$$

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Special Techniques in Electrostatics: Example 2



Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Define BC

- $V = 0$, at $y = 0$
- $V = 0$, at $y = a$
- $V = V_0$ at $x = b$
- $V = V_0$ at $x = -b$

$$V(x, y) = (Ae^{kx} + Be^{-kx})(C \sin ky + D \cos ky)$$

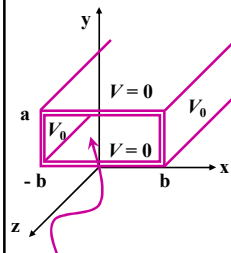
$A \neq 0$, as there is no extension for x to infinity

However, there is symmetry w.r.t. x : $V(-x, y) = V(x, y) \rightarrow A = B$

$$A(e^{kx} + e^{-kx}) = 2A \cosh kx$$

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Special Techniques in Electrostatics: Example 2



Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

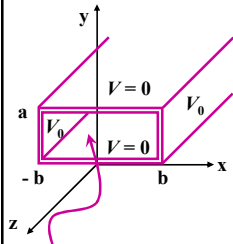
Define BC

- $V = 0$, at $y = 0$
- $V = 0$, at $y = a$
- $V = V_0$ at $x = b$
- $V = V_0$ at $x = -b$

$$V(x, y) = \cosh kx (C \sin ky + D \cos ky)$$

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Special Techniques in Electrostatics: Example 2

What is V inside?

Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

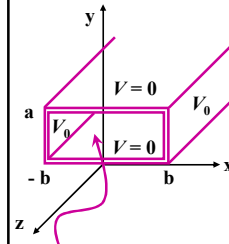
Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0$ at $x = b$
- iv. $V = V_0$ at $x = -b$

$$V(x, y) = \cosh kx C \sin ky \quad k = n\pi/a$$

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Special Techniques in Electrostatics: Example 2

What is V inside?

Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

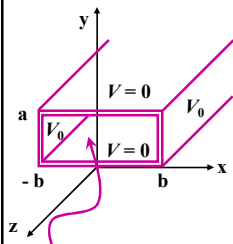
Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0$ at $x = b$
- iv. $V = V_0$ at $x = -b$

$$V(x, y) = C \cosh(n\pi x/a) \sin(n\pi y/a) \quad k = n\pi/a$$

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Special Techniques in Electrostatics: Example 2

What is V inside?

Symmetry

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = V_0$ at $x = b$
- iv. $V = V_0$ at $x = -b$

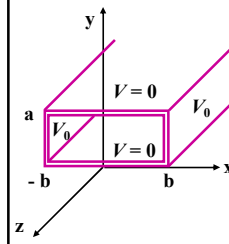
$$V(x, y) = C \cosh(n\pi x/a) \sin(n\pi y/a)$$

General Solution $V(x, y) = \sum_{n=1}^{\infty} C_n \cosh(n\pi x/a) \sin(n\pi y/a)$

$$V(b, y) = \sum_{n=1}^{\infty} C_n \cosh(n\pi b/a) \sin(n\pi y/a) = V_0$$

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Special Techniques in Electrostatics: Example 2



$$C_n \cosh(n\pi b/a) = \frac{2V_0}{a} \int_0^a \sin(n'\pi y/a) dy = \begin{cases} 0 & n \text{ being even} \\ \frac{4V_0}{n\pi} & n \text{ being odd} \end{cases}$$

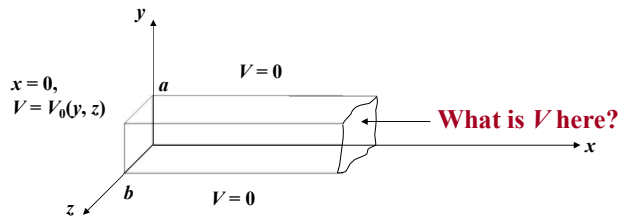
$$C_n = \frac{4V_0}{n\pi} \frac{1}{\cosh(n\pi b/a)} \quad n = 1, 3, 5, \dots$$

$$V(x, y) = \sum_{n=1}^{\infty} C_n \cosh(n\pi x/a) \sin(n\pi y/a)$$

$$V(x, y) = \frac{4V_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \frac{\cosh(n\pi x/a)}{\cosh(n\pi b/a)} \sin(n\pi y/a)$$

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Special Techniques in Electrostatics: Example 3



1. Look for symmetry $V = V(x, y, z)$

2. DE: LE as there is no charge in the region of interest

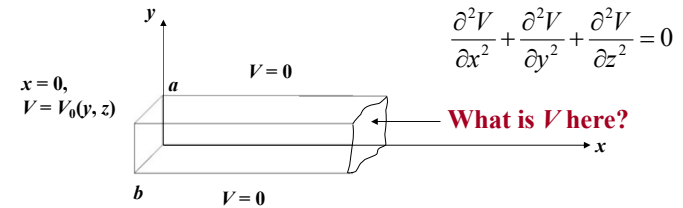
$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

3. Define BC

- i. $V = 0$, at $y = 0$
- ii. $V = 0$, at $y = a$
- iii. $V = 0$ at $z = 0$
- iv. $V = 0$ at $z = b$
- v. $V \rightarrow 0$, at $x \rightarrow \infty$
- vi. $V = V_0(y, z)$ at $x = 0$

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Special Techniques in Electrostatics: Example 3



$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

4. Variable separable method $V(x, y) = X(x)Y(y)Z(z)$

$$YZ \frac{d^2 X}{dx^2} + XZ \frac{d^2 Y}{dy^2} + XY \frac{d^2 Z}{dz^2} = 0$$

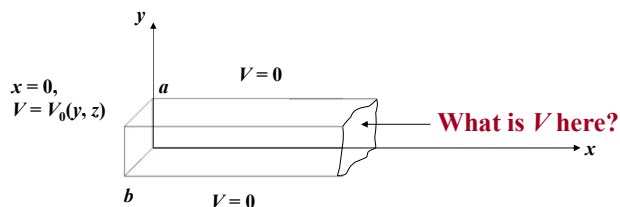
$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} + \frac{1}{Z} \frac{d^2 Z}{dz^2} = 0$$

$$f(x) + g(y) + h(z) = 0$$

$$C_1 + C_2 + C_3 = 0$$

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Special Techniques in Electrostatics: Example 3



$$\rightarrow \frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2 \quad \frac{1}{Z} \frac{d^2 Z}{dz^2} = C_3 \quad \text{PDE} \rightarrow \text{ODE}$$

As $V \rightarrow 0$, at $x \rightarrow \infty$, solution for x must be exponentially decaying

Along y and z , solution must be sinusoidal to meet BC

$\rightarrow C_1$ must +ve, C_2 and C_3 must be -ve

$$C_2 = -k^2 \quad C_3 = -l^2 \quad C_1 = k^2 + l^2$$

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Special Techniques in Electrostatics: Example 3

$$\frac{d^2 X}{dx^2} - (k^2 + l^2)X = 0 \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0 \quad \frac{d^2 Z}{dz^2} + l^2 Z = 0$$

$$X(x) = Ae^{\sqrt{k^2 + l^2}x} + Be^{-\sqrt{k^2 + l^2}x}$$

$$Y(y) = C \sin ky + D \cos ky$$

$$Z(z) = E \sin lz + F \cos lz$$

Apply BC i. $V = 0$, at $y = 0 \rightarrow D = 0$

iii. $V = 0$, at $z = 0 \rightarrow F = 0$

v. $V \rightarrow 0$ as $x \rightarrow \infty \rightarrow A = 0$

ii. $V = 0$ at $y = a \rightarrow k = n\pi/a \quad n = 1, 2, 3, \dots$

iv. $V = 0$, at $z = b \rightarrow l = m\pi/b \quad m = 1, 2, 3, \dots$

$$V(x, y, z) = Ce^{-x\sqrt{k^2 + l^2}} \sin(n\pi y/a) \sin(m\pi z/b)$$

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Special Techniques in Electrostatics: Example 3

$$\frac{d^2 X}{dx^2} - (k^2 + l^2)X = 0 \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0 \quad \frac{d^2 Z}{dz^2} + l^2 Z = 0$$

$$X(x) = Ae^{\sqrt{k^2 + l^2}x} + Be^{-\sqrt{k^2 + l^2}x}$$

$$Y(y) = C \sin ky + D \cos ky$$

$$Z(z) = E \sin lz + F \cos lz$$

Apply BC i. $V = 0$, at $y = 0 \rightarrow D = 0$

iii. $V = 0$, at $z = 0 \rightarrow F = 0$

v. $V \rightarrow 0$ as $x \rightarrow \infty \rightarrow A = 0$

ii. $V = 0$ at $y = a \rightarrow k = n\pi/a \quad n = 1, 2, 3, \dots$

iv. $V = 0$, at $z = b \rightarrow l = m\pi/b \quad m = 1, 2, 3, \dots$

General Solution

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y/a) \sin(m\pi z/b)$$

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Special Techniques in Electrostatics: Example 3

General Solution

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y/a) \sin(m\pi z/b)$$

Apply BC vi. $V = V_0(y, z)$, at $x = 0$

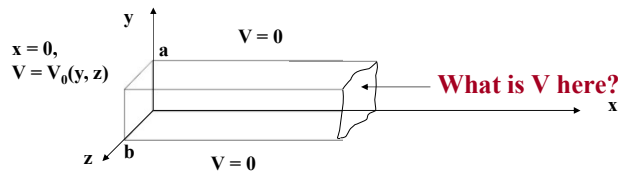
$$V(0, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \sin(n\pi y/a) \sin(m\pi z/b) = V_0(y, z)$$

To get C_{nm} , use Fourier Trick by multiplying $\sin(n'\pi y/a)$ and $\sin(m'\pi z/b)$ and integrate from $0 - a$ and $0 - b$

$$\begin{aligned} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \int_0^a \sin(n\pi y/a) \sin(n'\pi y/a) dy \int_0^b \sin(m\pi z/b) \sin(m'\pi z/b) dz \\ = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} \frac{a}{2} \delta_{nn'} \frac{b}{2} \delta_{mm'} = \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y/a) \sin(m'\pi z/b) dy dz \\ \rightarrow C_{n'm'} = \frac{4}{ab} \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y/a) \sin(m'\pi z/b) dy dz \end{aligned}$$

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Special Techniques in Electrostatics: Example 3



Unique Solution

$$V(x, y, z) = \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y/a) \sin(m\pi z/b)$$

$$C_{n'm'} = \frac{4}{ab} \int_0^a \int_0^b V_0(y, z) \sin(n'\pi y/a) \sin(m'\pi z/b) dy dz$$

Special case: $V_0(y, z) = V_0$

$$C_{n'm'} = \frac{4V_0}{ab} \int_0^a \sin(n'\pi y/a) dy \int_0^b \sin(m'\pi z/b) dz = \begin{cases} 0 & n, m \text{ being even} \\ \frac{16V_0}{nm\pi^2} & n, m \text{ being odd} \end{cases}$$

$$V(x, y, z) = \frac{16V_0}{\pi^2} \sum_{n=1,3,5,\dots}^{\infty} \sum_{m=1,3,5,\dots}^{\infty} \frac{1}{nm} e^{-x\pi\sqrt{(n/a)^2 + (m/b)^2}} \sin(n\pi y/a) \sin(m\pi z/b)$$

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